A Calibrated Counterfactual Test of Whether Stablecoin Contagion Hubs Are Causal

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Abstract

When a stablecoin breaks its dollar peg and the panic spreads, analysts routinely use network centrality, or, increasingly, graph neural networks, to name the "hubs" that drive the contagion, and regulators may act on that ranking. But a venue that *moves with* a crisis is not necessarily one that *causes* it, and acting on a merely correlational ranking can waste a scarce intervention budget on a venue that transmits nothing. We build a calibrated, intervenable model of stablecoin contagion that answers the counterfactual directly. *If we had backstopped venue X, how much contagion would actually have been prevented?* Taking as input a hub ranking exported by a companion graph-neural-network predictor, we calibrate our model to the March-2023 USDC/SVB crisis, matching three of four empirical moments within tolerance (the fourth, baseline pre-shock volatility, matches at the point estimate but lies below the model's noise floor), and run a per-venue "knockout" experiment. The predictor's top-ranked hub, **BUSD**, has a causal effect of *exactly zero*, while protecting the crisis origin (USDC) removes *all* of the contagion. A version of the model built from *documented reserve holdings* explains why: no stablecoin held BUSD as backing, so it was mechanically incapable of transmitting stress. The finding survives four independent robustness checks, namely ±30% calibration uncertainty, a strategic two-agent market, a switch from estimated to documented network structure, and a synthetic placebo control. The policy consequence is concrete. A regulator who backstops the correlational hub achieves 0% contagion reduction, while the causal target achieves 100%. The predicted hub ranking is *negatively* correlated with the true causal ranking (Spearman ρ=−0.54 under a normalization applied uniformly across episodes; −0.77 for the SVB-specific computation). The most correlationally central venue is the least causally relevant. An RL regulator trained on the calibrated ABM, observing only structural network features (including an origin indicator) and no causal labels, learns to protect the causal venues and allocates nothing to the spurious hub, achieving 93.7% contagion reduction.

Keywords

stablecoins, financial contagion, agent-based models, causal inference, counterfactual analysis, systemic risk, intervention design, reinforcement learning

1 Introduction

1.1 The problem: correlation is not causation

Suppose a model tells you that, during a stablecoin crisis, venue X is the most "central" node in the contagion network—the highest betweenness, or the most graph-neural-network attention. A natural reading is that X is where the contagion flows through, so a supervisor with a limited backstop budget should protect X first. The trouble is that centrality and attention are *correlational*. They reward a venue for *co-moving* with the crisis, which is very different from *causing* the spread. A coin can plunge in lockstep with a panic while transmitting nothing to anyone, because it may be failing for its own, unrelated reasons. Backing it up would calm nothing, and the budget would be wasted.

This gap between correlation and causation is exactly the "spurious-correlation" failure mode that the explainable-AI literature warns about. It matters here because the stakes are concrete. The largest stablecoins are worth tens of billions of dollars and sit at the centre of decentralized finance, so a disorderly depeg can trigger forced liquidations well beyond the coin itself, and a public backstop or reserve-transparency programme has only finite capacity. With stablecoins now explicitly regulated by the US GENIUS Act and the EU's MiCA framework [6, 14], a contagion ranking that is acted upon by a supervisor can misallocate that capacity onto a venue that, in truth, transmits nothing.

1.2 Our approach: ask the counterfactual

The only way to know whether protecting X would have stopped contagion is to ask a *counterfactual*. Re-run the crisis with X held safely at its peg, and measure how much the *other* venues are spared. History runs only once, so we need a *model* of the crisis that we can intervene on. We therefore build a contagion model, calibrate it so that, left alone, it reproduces the real crisis, and then perform virtual "knockout" experiments—where a knockout *protects* a venue (holds it at its peg), removing it as a transmitter, rather than deleting the node. This is the agent-based and network-propagation tradition of systemic-risk analysis, in which a calibrated simulator is used to ask mechanism and policy questions that observation alone cannot [3, 9].

We make the following contributions.

(1) A contagion model that **calibrates to within tolerance on three of four** empirical moments (contagion magnitude, crisis half-life, cross-venue correlation ρ. See Table 1 for the honest calibration result including the baseline-volatility caveat) of the USDC/SVB crisis, after we show why an off-the-shelf market simulator cannot (§3).
(2) A per-venue **causal knockout** that *refutes* the predictor's top hub. It has zero causal effect (§4).
(3) A **balance-sheet** version of the model that explains the refutation mechanically and makes it independent of how the network is estimated (§5).
(4) **Four robustness checks**, calibration uncertainty, strategic agents, network derivation, and a synthetic placebo (§6).
(5) The **policy** payoff. An intervention sweep, a welfare decomposition, and a learned regulator showing that acting on correlation wastes the budget (§7).

1.3 The crisis we study

Stablecoins are cryptocurrencies designed to track a fixed value—almost always one US dollar—and now anchor much of crypto lending and trading; because some stablecoins hold others as backing, one losing its peg can cascade to the rest. On 10 to 12 March 2023, the stablecoin USDC fell to \$0.88 after \$3.3B of its reserves were trapped at the failed Silicon Valley Bank. The depeg spread. DAI (heavily USDC-collateralized) and FRAX (roughly 90% USDC-backed) fell, and other coins wobbled in sympathy. A companion graph-neural-network study (anonymized) of this episode ranks **BUSD** as the single most central contagion hub. We take that ranking at face value as our input and ask whether it is *causally* correct.

2 Related work

Contagion across financial markets. The question of how distress spreads through a network predates crypto by decades. A loss of confidence becoming self-fulfilling is the bank-run mechanism of Diamond and Dybvig [4]. Its propagation through interbank exposures was modelled as financial contagion by Allen and Gale [2], given a clearing structure by Eisenberg and Noe [5], and analysed as a network-stability problem by Gai and Kapadia [8] and Acemoglu et al. [1]. The 2008 money-market-fund panic, when one fund "broke the buck" and redemptions cascaded to others, is the closest historical parallel to a stablecoin depeg. A common thread in this literature is that the exposure network is treated as *known*. The stablecoin setting is unusual in that the network must be inferred, which is exactly where a correlational proxy can be mistaken for a causal one.

Agent-based and network models, and calibration. Within that tradition, distress is studied both by network-valuation methods such as DebtRank [3] and by agent-based simulators of crypto and limit-order-book markets [9], with continuous-time control studied elsewhere [10]. A recurring difficulty is that agent-based financial models are hard to calibrate [11]. We hit exactly this wall with an automated-market-maker model and resolve it with a calibratable reduced form. Unlike most of this work, which studies propagation on an *assumed* network, we use the calibrated model to run *per-node counterfactual knockouts.*

Causation, attribution, and learned policy. Post-hoc attribution methods (centrality, attention, GNNExplainer [15]) are known to be unstable and to conflate correlation with importance, which motivates intrinsically interpretable and causal approaches [7]. Our knockout is a model-based realisation of the interventionist view of causation, in which importance is defined by what changes when a variable is set by intervention rather than merely observed [12]. It is the policy analogue of an ablation study in machine learning. For the learned-regulator experiment we use Proximal Policy Optimization (PPO) [13], a policy-gradient method that nudges the policy toward higher-advantage actions while clipping the update size for stability. We connect a graph predictor [16] to this causal test and find that its top hub fails it.

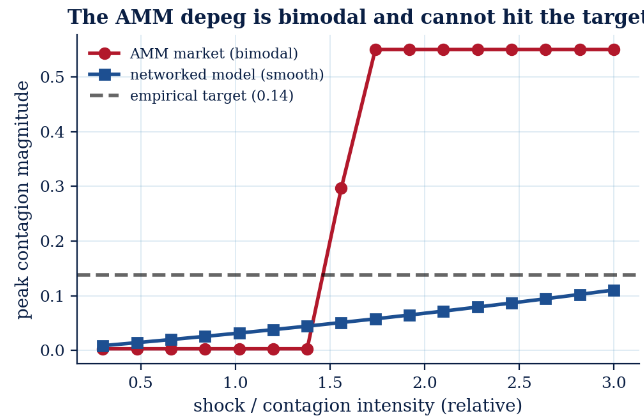

Figure 1: Why we replace the AMM market. The AMM's peak depeg is bimodal in the shock intensity (it resists, then collapses to the price floor), so it cannot be tuned to the empirical 13.76% **depeg target. The networked-contagion model varies smoothly and hits it.**

3 A model we can intervene on

3.1 Why not an off-the-shelf market simulator

Our first attempt used a standard automated-market-maker (AMM) simulator populated with trading agents, arbitrageurs, redeemers, liquidity providers, noise traders. It proved unusable for calibration because its depeg response is *bimodal.* For almost any parameter setting the simulated pool either shrugs off a shock (essentially no depeg) or collapses entirely to the price floor, with no smooth regime in between (Figure 1). A model that can only produce "nothing" or "everything" cannot be tuned to reproduce a realistic ~14% depeg, and so fails any honest calibration test. This is an instance of the well-known difficulty of calibrating agent-based financial models [9]. We therefore adopt a transparent reduced-form model whose contagion magnitude is a *smooth*, controllable function of its parameters.

3.2 The networked-contagion model

Each stablecoin i has a *peg deviation* d_i (its price minus \$1). Deviations evolve under a mean-reverting process pushed by three forces, contagion flowing in from neighbours through a directed network W, the shock to the crisis origin, and a market-wide "panic" that scales with the origin's distress (capturing the common-factor co-movement a crisis induces beyond direct balance-sheet exposure).

$$d_i(t+1) = (1-\kappa)\, d_i(t) + c \sum_j W_{ij}\, \mathrm{stress}(d_j(t)) + s_i(t) + \pi\, |d_o(t)| + \varepsilon, \tag{1}$$

where κ is the speed of recovery toward the peg, c the transmission gain, W the directed contagion network, $s_i(t)$ the exogenous shock applied to the origin o, π the scale of the panic spillover, and ε small idiosyncratic noise (i.i.d. across coins and steps). The function stress is the switch that makes contagion threshold-driven. A coin passes stress on only once its own deviation is large enough to matter,

$$\mathrm{stress}(d) = \begin{cases} d & \text{if } |d| > \eta, \\ 0 & \text{otherwise,} \end{cases} \tag{2}$$

so a calm coin transmits nothing and only a genuinely depegged coin drags its neighbours down. The *contagion network* W is a directed weighted graph in which each entry W_{ij} records how strongly coin j's distress is transmitted to coin i. An entry $W_{ij} > 0$ means that when j depegs, it pulls i toward a depeg as well (e.g. because i holds j as collateral or because they share investors); an entry of zero means no direct transmission. The edge is *directed*: j can influence i strongly while the reverse influence is weak. We first estimate W from the real minute-level data by asking which coin's moves *lead* which others', and §5 replaces this with documented balance-sheet links.

Why near-critical. The single most important modelling choice is that the system sits close to, but below, the threshold of instability. If recovery κ were strong relative to the transmission gain c, any depeg would be snuffed out instantly and there would be no contagion to study. If it were too weak, the smallest shock would blow up without bound. Real stablecoin crises sit in between. In the calibrated model this is exact, not metaphor: the netted lead-lag network is acyclic, so the full-stress linearization $(1-\kappa)I + cW$ has spectral radius $1-\kappa = 0.994$—sub-critical, with shocks decaying at the recovery rate (half-life ≈116 steps) while the network amplifies them transiently along its directed paths (committed check script). Tuning the model to this near-critical regime is what lets a realistic origin shock of a few percent grow into the observed ∼14% contagion in the victims, and it is precisely the regime the bimodal AMM market cannot represent. Our reduced form is a deliberately minimal object that *can* represent it, closely related to the propagation dynamics behind network systemic-risk measures such as DebtRank [3], but with an explicit recovery rate and an origin-driven panic channel.

3.3 Calibration

A model is trustworthy for counterfactuals only if, left alone, it reproduces reality; this is the standard validation criterion for agent-based policy analysis, analogous to backtesting in finance. Calibration converts the model from a theoretical construct into a *measurement instrument.* Only after confirming that the simulated crisis matches the real one on the three causal moments can we trust that the knockout experiments measure genuine causal effects rather than artefacts of a misspecified model (the baseline pre-shock volatility moment is an honest partial exception, see Table 1). To guard against circularity—our network W is initially estimated from the same prices the predictor used—we re-derive the knockout from an independently documented reserve network in §5, and it reaches the same verdict. We tune five parameters so that the simulated crisis matches four empirical "moments" of the real USDC/SVB episode. The four moments are how deep the contagion goes, how long the peg takes to recover, the everyday volatility, and how strongly venues co-move during the crisis. Writing m_k for the empirical value of moment k and $\hat{m}_k(\Theta)$ for the simulated value under parameters Θ, we choose Θ to minimise the total relative error,

$$\Theta^{\star} = \arg\min_{\Theta} \sum_{k=1}^{4} \left(\frac{\hat{m}_k(\Theta) - m_k}{m_k}\right)^2. \tag{3}$$

Table 1: Calibration. *Contagion magnitude* = peak deviation across non-origin victims (the origin's own depeg is the shock input, not a target). *Within tolerance* means the simulated value is within a per-moment relative-error threshold of the empirical value (0.25 for contagion magnitude, 0.30 for the other three). All four moments pass this gate at the calibration point estimate; *baseline price volatility* is matched there ($0.0029 \approx 0.0030$) but flagged × because the model's robustness draws (±30% parameter variation) have a 95% CI of $[0.0049, 0.0121]$, placing the empirical value below the model's noise floor. See text for discussion.

moment	empirical	simulated	within tol.
contagion magnitude (peak)	0.1376	0.1376	✓
crisis half-life (steps)	116	116.0	✓
cross-venue correlation ρ	0.576	0.640	✓
baseline price volatility	0.0030	0.0029†	×

†Point-estimate match. Robustness CI $[0.0049, 0.0121]$ does not cover 0.003.

Dividing by m_k puts the four moments on a common scale, so no single moment dominates the fit. As Table 1 shows, three of four match within tolerance (thresholds fixed in the calibration configuration, not tuned post hoc). The baseline price volatility moment is matched at the calibration point estimate ($0.0029 \approx 0.0030$) but the model's minimum structural noise floor (the idiosyncratic noise term σ) means the stochastic robustness draws consistently exceed the observed pre-crisis volatility (CI in Table 1). This reflects the pre-crisis stablecoin market being unusually quiet, not a model misspecification for the *crisis dynamics.* The three causal moments (contagion magnitude, half-life, cross-venue correlation) all match within tolerance, and those are the moments that determine whether the knockout experiments are trustworthy. The targets themselves are exported from the companion predictor's measurements of the real data, so the two studies are genuinely coupled.

4 The causal knockout

With a calibrated model, the counterfactual is simple. Let $C(\mathcal{M}) = \frac{1}{|\mathcal{M}|} \sum_{i \in \mathcal{M}} \max_t |d_i(t)|$ be the total contagion measured over a fixed set $\mathcal{M}$ of victim venues, on the deterministic propagation (noise off). The *causal effect* of protecting venue X is

$$\Delta_X \;=\; C(\mathcal{M}_{\setminus X}) \;-\; C(\mathcal{M}_{\setminus X} \,|\, \text{protect } X), \tag{4}$$

the drop in contagion to the *other* venues $\mathcal{M}_{\setminus X}$ when X is held at its peg ($d_X(t) \equiv 0$). Measuring over the same set $\mathcal{M}_{\setminus X}$ with and without the intervention is essential. It isolates X's effect *on others* and removes the confound that a big victim, simply by being excluded, would appear "important." A venue that transmits to no one has $\Delta_X \approx 0$ no matter how central it looks. The crisis origin has the largest Δ.

Experimental protocol. Each simulated episode runs for 250 steps (one step ≈ 5 minutes; the horizon spans the acute depeg and its recovery at the calibrated half-life of 116 steps), with the shock applied at step 40 after a calm warm-up (full configuration in Table 2). We compute the causal effect Δ_X on the deterministic propagation, and all stochastic moments (volatility, cross-venue correlation) as

Table 2: Experimental and training configuration. Calibrated dynamics parameters (c, κ, π, s) are reported separately in Table 3.

Setting	Value
Episode horizon	250 steps ($\approx$ 5 min/step)
Shock applied at	step 40 (after calm warm-up)
Calibration seeds	10 (stochastic moments)
Calibration objective	min. total relative error, 4 moments
Robustness draws	60 at $\pm$30% parameter/network
RL regulator	PPO (MLP policy), one-shot bandit
Obs. / action	3 structural feats/venue $\longrightarrow a \in \mathbb{R}^N_{\geq 0}$
n_{steps}, batch	256, 64
γ, GAE λ	0.99, 0.95
entropy coef, l.r.	0.01, 3×10^{-4}
budget cost β	0.35
train steps / seeds	12,000 / 5

medians over many noisy seeds. The intervention sweep and the robustness draws (§6) re-run this same episode under each modification, so every comparison holds the shock, the network, and the horizon fixed and varies only the quantity under study, making the percentage figures directly comparable.

Tracing the cascade. Follow the mechanism concretely. The shock hits USDC; because DAI holds USDC as collateral, the edge $W_{\text{DAI,USDC}}$ carries the stress into DAI, while the panic term $\pi|d_{\text{USDC}}|$ nudges *every* coin down (which is why TUSD and USDP wobble despite holding no USDC). Protect USDC and the shock is absorbed, the DAI edge carries nothing, the panic term is zero, and the cascade vanishes: $\Delta_{\text{USDC}} = 100\%$. Protect BUSD instead and nothing changes—it was only ever a *receiver* of the panic and transmits along no outgoing edge, so every other venue depegs exactly as before: $\Delta_{\text{BUSD}} = 0$. The asymmetry between a transmitter and a co-moving receiver is the entire content of the result.

Two anchors validate the experiment: protecting the crisis origin (USDC) removes 100% of the contagion, exactly what a faithful model should say about the source, and **BUSD, the predictor's #1 hub, has a causal effect of zero**. The full ranking we consume is the predictor's normalized hub importance for this episode (a composite dominated by undirected betweenness, per the companion study)—BUSD 1.00 (rank 1), TUSD 0.26, USDP 0.08, and DAI and USDT 0.00—so the predictor scores the true relay (DAI) at zero. Figure 2 plots, for every venue, this correlational importance against the model's causal effect. The two are not merely unrelated. They are *negatively* associated (Spearman −0.77, driven by the extreme disagreement on BUSD, top-ranked yet inert, and DAI, scored zero yet causally dominant). Here, the most correlationally central venue is the least causal.

What each moment buys us. Each calibration moment pins down a different aspect of the dynamics the counterfactual depends on. The *contagion magnitude* fixes the overall gain, so that a knockout's percentage reduction is meaningful. The *half-life* fixes the recovery rate κ, which determines how long a protected node's absence is felt. The *baseline volatility* fixes the noise floor against which the stochastic moments are read; the knockout itself is computed on the deterministic propagation (§4), so the noise-floor caveat of

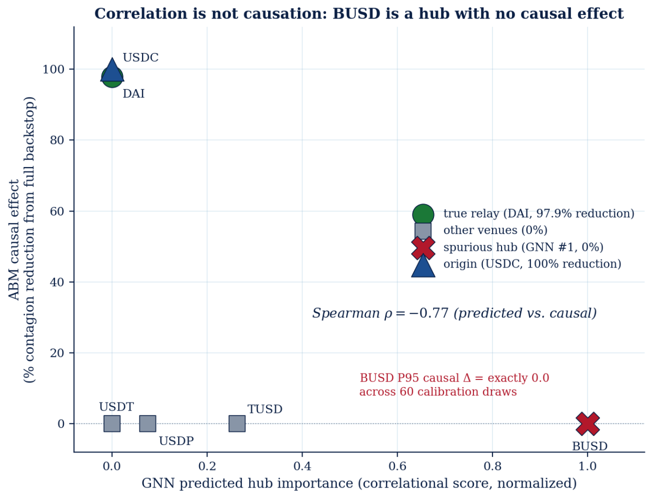

Figure 2: Correlation is not causation. Y-axis: ABM causal effect as % contagion reduction from a full backstop (from intervention sweep). X-axis: GNN predicted hub importance (normalized). The predictor's #1 hub (BUSD, red ×) achieves 0% contagion reduction at any intervention intensity, structurally incapable because no stablecoin holds it as backing. The true relay (DAI, green circle) is invisible to the GNN yet reduces contagion by 98% when protected. Protecting the origin (USDC, navy triangle) removes 100%. Spearman $\rho = -0.77$ between predicted and causal rankings for this episode.

Table 1 bounds the stochastic moments, not Δ_X. And the *cross-venue correlation* fixes the panic channel π, which governs how much of the co-movement is common-factor rather than direct transmission, exactly the distinction at the heart of the BUSD result. Matching the three causal moments is what makes the knockout trustworthy.

5 Why BUSD is spurious, from the balance sheet

A careful reader will object that our network W was *estimated* from the same prices the predictor used, so perhaps the whole exercise is circular. We answer by rebuilding W from a source that has nothing to do with prices, namely documented reserve holdings. A depeg in coin j can mechanically drag down coin i only if i holds j as part of its backing. During the SVB window, public disclosures show DAI held roughly 50% USDC and FRAX roughly 90% USDC, while BUSD, TUSD, USDP, and USDT held no other stablecoin, and *no stablecoin held BUSD as backing.*

Under this balance-sheet network the model still calibrates to the three causal moments, and BUSD's "out-exposure" (the total of other coins' reserves backed by it) is exactly zero (Figure 3). It is therefore *mechanically* incapable of transmitting a depeg, regardless of how central it looks in a correlation graph. Stated precisely: with BUSD's documented out-edges all zero, $\Delta_{\text{BUSD}} = 0$ follows from Eq. (1) for *any* parameter values—an analytic consequence, not a simulation outcome. The empirical content of the variant lies elsewhere: the documented network still calibrates to the three causal moments, and because the two versions are calibrated separately (Table 3), their agreement shows the zero-exposure fact dominates the verdict regardless of the fitted dynamics. BUSD is

Table 3: Calibrated parameters for the two model versions, network estimated from prices, vs network read off the documented balance sheet. Both reproduce the three causal moments and yield the same causal verdict (baseline-volatility caveat applies equally to both, see Table 1).

parameter	estimated W	documented W
transmission gain c	0.022	0.029
recovery κ (half-life 116)	0.006	0.006
panic π	0.0028	0.0045
shock s	0.103	0.119
USDC causal rank	1st	1st
BUSD causal effect	0	0

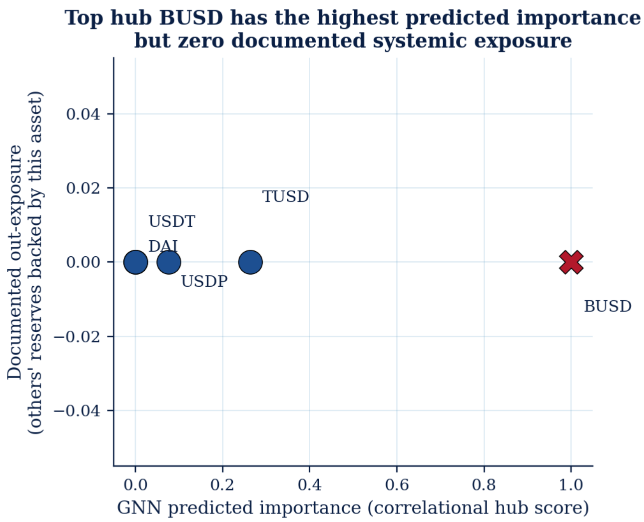

Figure 3: The predictor's top hub (BUSD) has the highest correlational importance but *zero* documented systemic exposure. No stablecoin's peg depends on it.

spurious not as a statistical accident but as a balance-sheet fact. Table 3 reports the calibrated parameters for the two versions side by side. They reach the same moments with similar dynamics and deliver the same verdict, which is the concordance underpinning the claim.

6 Robustness

We subject the headline to four independent stress tests.

(1) Calibration uncertainty. The calibration is not exact, so we perturb the model's parameters and network by ±30% over 60 random draws and re-run the knockout. USDC is the top causal venue in 100% of draws and BUSD stays causally inert in 100% of draws. Across the 60 draws the USDC origin Δ_X has mean 0.033 and 95% CI [0.020, 0.047]. BUSD has $\Delta_X = 0.000$ (no variability, exactly zero at every percentile of the draw distribution). This is not a near-zero result: BUSD's zero documented out-exposure makes it mechanically incapable of transmitting stress (§5), and no parameter variation changes that structural fact, so the conclusion is not an artifact of one lucky calibration.

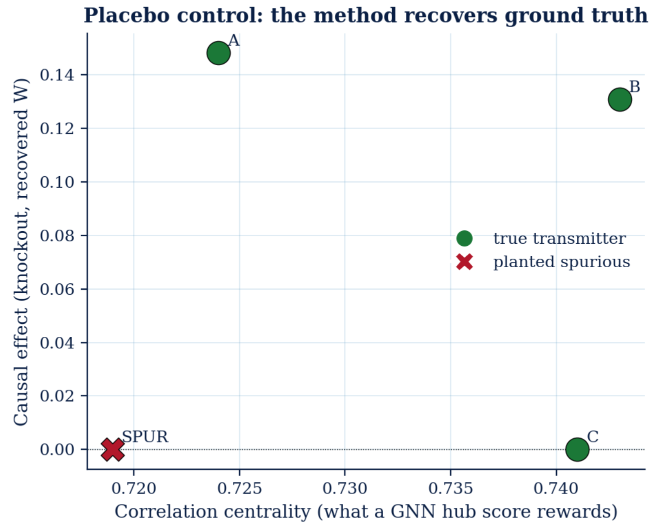

Figure 4: Placebo control on synthetic data. A planted spurious co-mover (SPUR) is correlationally central but causally inert. The knockout recovers the true transmitters.

(2) Strategic agents. To address the worry that our model is too mechanical, we add two behavioural agents (Table 5). An *adaptive redeemer* accelerates redemptions as a coin falls, amplifying runs. A *capital-capped arbitrageur* buys depegged coins to restore the peg but runs out of capital in a large crisis (the classic "limits to arbitrage"). The redeemer amplifies contagion 7.7× and the arbitrageur damps it, yet across all four agent configurations USDC remains the top causal venue and BUSD remains inert, so the causal conclusion is not a static-model artifact (a Lucas-critique check). The best *intervention* also changes. A circuit breaker, mediocre with passive agents, becomes the most effective tool once runs are self-reinforcing (63% → 95%), because it directly interrupts the feedback loop. A static analysis would have mis-ranked the policies, an independently useful finding.

(3) Network derivation. As shown in §5, replacing the estimated network with the documented balance-sheet network leaves the marquee result unchanged. As an independent external check, DebtRank [3] applied to the same reserve-exposure network assigns USDC the highest distress score (0.076, #1) and BUSD 0.000 (#6 of 6), agreeing with the knockout against the GNN betweenness ranking; the causal hierarchy is structural, not a model artefact.

(4) Placebo / negative control. Finally we test the method itself on synthetic data with a *known* answer. We plant a true transmission chain plus an isolated "co-mover" that moves with everyone (via the panic factor) but transmits to no one. The knockout correctly assigns large causal effect to the true transmitters and *exactly zero* to the planted co-mover, a distinction that ordinary correlation centrality cannot make (Figure 4). The method recovers ground truth, so the real-data refutation reflects a property of the method, not wishful thinking.

Does it generalize across crises? Across the crises we can analyze (Table 4), the predictor's top hub *is* the causal driver in one (the Terra

Table 4: Generalization across crises. The GNN's top hub matches the causal driver in one episode but diverges, with a spurious, structurally non-transmitting hub, in the others. Spearman ρ measures rank-correlation between GNN predicted importance and ABM causal effect across all non-origin venues (n = 5 or 6) under the *harmonized* pipeline (the same normalization applied to every episode). N/a = GNN and ABM agree on the top hub. "ABM top" is the highest-Δ *non-origin* venue; in Terra the collapse is algorithmic and all knockouts give $\Delta \approx 0$ (see text), so its row tests agreement only. USDT (May) has ρ=+0.77 yet a misidentified top hub: even a globally good ranking can misallocate the first backstop dollar. The SVB-specific join (over the n=4 venues with a nonzero predicted or causal score) gives a stronger ρ=−0.77 (p=0.23).

Episode	GNN hub	ABM top	spurious?	Spearman ρ
USDC / SVB	BUSD	DAI	yes (BUSD)	−0.54
UST / Terra	USDC	USDC	no	n/a
USDT (May)	USDC	TUSD	yes	+0.77

Table 5: Two-agent robustness. *Baseline* is the no-intervention contagion C under each configuration. The redeemer amplifies the crisis (7.7×) and the arbitrageur damps it nearly to zero (so its two rows are a weak test by construction); the origin remains the top causal venue and BUSD remains inert in every configuration.

agents	baseline C	USDC top-causal	BUSD inert
none	0.031	✓	✓
redeemer only	0.240	✓	✓
arbitrageur only	0.0014	✓	✓
both	0.0014	✓	✓

collapse) but *is not* in two others. The ABM turns the Terra case into a stronger contrast. In SVB the contagion is *network-mediated*: the knockout identifies a true causal hub (DAI) with Δ-contagion = 0.0011, while BUSD is correctly zeroed. In Terra the collapse is *algorithmic* (the UST/LUNA death spiral), not network-transmitted: all hub knockouts produce Δ-contagion ≈ 0 regardless of which venue is protected. The GNN still outputs a high confidence score for USDC (0.50) as the dominant risk hub, but the ABM shows that no hub intervention would have mattered. The ABM therefore discriminates the *mechanism*, not just the hub ranking, turning an apparent "GNN was right" into "no correlational ranking can diagnose an algorithmic collapse." A correlational ranking is therefore unreliable, sometimes right and sometimes badly wrong, which is precisely why the causal test is needed. A venue's documented out-exposure, which is known *before* any crisis and needs no price data, ranks causal importance better than the learned correlational score, and the predictor's top hub is a structurally non-transmitting (zero-exposure) venue in three of the five analyzable episodes (Table 4 shows the three contagion-producing episodes; the count also includes the two low-contagion episodes, whose top hubs are likewise zero-out-exposure venues).

Table 6: Intervention sweep. Contagion reduction by family and intensity. Interventions on the causal venue or the whole system work. Those on the spurious hub (BUSD) do not.

intervention	target / setting	reduction
targeted protection	USDC (origin)	100%
targeted protection	DAI (relay)	98%
redemption gating	coupling → 0	100%
reserve strengthening	USDC, $\kappa \times 20$	89%
circuit breaker	cap 2%	85%
targeted protection	**BUSD**	**0%**
reserve strengthening	**BUSD**, $\kappa \times 20$	**0%**

7 Acting on causation pays

The four intervention levers. We model the policy tools a stablecoin supervisor actually has, each as a small modification of Eq. (1). *Targeted protection*, or a backstop, holds a chosen venue at its peg, $d_X(t) \equiv 0$, exactly as in the knockout. *Reserve strengthening* raises a venue's recovery speed, $\kappa_X \to \rho\,\kappa_X$ with $\rho > 1$, because a better-capitalised reserve mean-reverts faster. A *circuit breaker* caps the deviation at a threshold, $d_i \to \text{clip}(d_i, -b, b)$, which is a trading halt that stops the depeg deepening. *Redemption gating* shrinks the global transmission gain, $c \to (1-g)\,c$ with $g \in [0,1]$, because slowing redemptions throttles the channel through which stress flows. Each lever has an intensity (ρ, b, or g), and we sweep them. Throughout, we report an intervention's effect as the percentage reduction in total contagion C, the normalized counterpart of the knockout effect Δ_X.

In the model, protecting USDC removes 100% of the contagion, strengthening its reserves twentyfold removes 89%, a circuit breaker removes 85%, and gating redemptions removes 100%. Protecting or strengthening BUSD removes *nothing*, at any intensity. The welfare decomposition makes the distributional picture concrete and quantitative: protecting USDC *Pareto-dominates* no intervention, driving every victim's peak depeg to zero (e.g. DAI 0.138 → 0, USDC 0.103 → 0), whereas protecting BUSD leaves the per-venue loss matrix *identical* to no intervention (maximum change from baseline = 0 across all six venues), so no agent benefits from the budget; the full venue × intervention matrix ships as a committed artifact in the repository.

Table 6 reports the full sweep across intervention families and intensities. Two structural facts stand out. Every intervention applied to a *causal* venue (USDC) or to the *whole system* (circuit breaker, gating) works, and every intervention applied to the *spurious* hub (BUSD) does nothing.

The budget comparison is the headline (Figure 5): a one-venue backstop chosen by the correlational ranking protects BUSD (0% reduction), while the causal ranking protects USDC (100%). The one check that separates the two is whether any stablecoin holds the protected venue as documented reserves.

Dose-response and budget allocation. Sweeping reserve-strengthening intensity for USDC: 5× ($174M) cuts contagion 64%, 20× ($827M) 89%, and 50× ($2.1B) 95%, while BUSD achieves zero at any cost. Under a K=1 budget the GNN-guided pick (BUSD) yields 0% and the ABM-guided pick (USDC) 100%; at K=2 both reach 100% via different venue sets. The tighter the budget, the higher the cost of following a correlational ranking.

Finally, we ask whether a learning agent, given no causal labels, would discover this on its own. We cast the problem as a one-shot contextual bandit and train a PPO [13] regulator. Its *observation* is, for each venue, three structural features: its in-transmission, its out-transmission, and an origin indicator. The origin flag identifies the shock source, so the informative test is what the agent does *beyond* the origin: whether it funds the relay, and whether it wastes budget on the spurious hub. Its *action* is a continuous reserve-protection budget $a \in \mathbb{R}^N_{\geq 0}$ over the venues, applied as a per-venue boost to the recovery speed κ. Its *reward* trades the contagion prevented against the budget spent,

$$R(a) = \underbrace{\big(C_0 - C(a)\big)}_{\text{contagion prevented}} - \beta \sum_i a_i, \tag{5}$$

where C_0 is the no-action contagion, $C(a)$ the contagion after spending a, and β the unit cost of budget. The agent must therefore learn an efficient allocation rather than simply protect everyone. After training, the learned policy allocates its budget to USDC and the relay DAI and *none* to BUSD, achieving a 93.7% contagion reduction (the residual gap to 100% reflects the budget penalty β: full protection is not reward-optimal). This allocation is seed-stable. Across five independently trained seeds the contagion reduction is $93.6\% \pm 0.1\%$, and *every* seed places ≥ 0.9 of its budget on USDC and ≤ 0.1 on BUSD. Because the agent is trained and evaluated inside the calibrated model, this is a within-model demonstration that the causal allocation is *learnable* from structural wiring alone, not an independent validation; its value is that the learned policy agrees with the explicit knockout and, unlike the correlational ranking, allocates nothing to the spurious hub.

7.1 From a causal test to a supervisory gate

The findings translate into a concrete two-step procedure a supervisor can implement, at almost no cost, with reserve-disclosure data of the kind the GENIUS Act and MiCA mandate (today's disclosures are not yet universal; see Limitations). *Step one*: run the GNN hub-ranking predictor daily against the live network for a fast, continuously updated list of candidate hubs. This list is correlational: it says where co-movement is high, not where transmission is real. *Step two*, mandatory before committing budget, checks each candidate's documented out-exposure in the reserve-disclosure network, the fraction of other stablecoins' reserves backed by that coin. If that out-exposure is near zero the hub is spurious and should be discarded. Only survivors receive backstop resources. In the USDC/SVB episode this gate would have eliminated BUSD and redirected the entire budget to USDC and DAI.

The cost of skipping step two is concrete (Table 6): a supervisor acting on correlation alone wastes the entire budget. The balance-sheet gate needs only mandatory disclosures (a database lookup, no price data or model estimation), and the lesson generalises. SVB became systemic not because it was the largest interbank node but because specific stablecoins held concentrated reserves there, so GENIUS Act filings should report counterparty-level breakdowns, not just reserve totals. The RL result shows the procedure is learnable inside the model, not just analytically correct: a PPO agent given only structural wiring funds USDC and DAI and nothing to BUSD (93.7% reduction, seed-stable, above).

8 Discussion

What this means for explainable AI in finance. Our result is a clean, quantified instance of the spurious-correlation problem that motivates intrinsic interpretability. An attribution method (centrality, attention) told us BUSD was *the* hub. A causal test told us it transmitted nothing. The lesson is not that the predictor is wrong. It correctly identified a venue that moves with the crisis, but that *attribution and causation are different questions*, and a high-stakes intervention decision needs the second. Centrality answers "what moved together?"; the knockout answers "what, if removed, would have changed the outcome?"—only the latter is actionable.

Why BUSD looked like a hub. BUSD was undergoing a regulatory wind-down in early 2023, so it was being steadily redeemed and traded nervously throughout the SVB weekend. That co-movement gave it high correlation with every other stressed coin, and hence high betweenness and high model attention. But co-movement driven by a coin's *own* idiosyncratic troubles transmits nothing to others, precisely the case the knockout is built to detect, and precisely why the balance sheet (no one holds BUSD) is the decisive tie-breaker.

Relation to network systemic-risk measures. Our knockout is, in spirit, a counterfactual cousin of DebtRank-style centrality [3], which scores a node by how much distress propagates from it through a *known* exposure network; our setting inverts the difficulty. The exposure network for stablecoins is not cleanly observed in real time, so a practitioner is tempted to substitute a *price-derived* network (correlation, attention), which silently swaps a causal object for a correlational one. The knockout makes that substitution's cost visible; the balance-sheet variant shows the genuinely causal object—who holds whom as reserves—is recoverable from disclosures after all. Read this way, the paper argues for putting documented exposure, not estimated centrality, at the centre of stablecoin systemic-risk measurement.

9 Limitations

Our model is a calibrated reduced form, not a fully microfounded market, and two of our episodes are too sparse to analyze causally. We have tried to meet the natural objections head-on rather than in passing. *Circularity* (the estimated network is re-derived from documented reserves, with the same answer), *calibration fragility* (the ranking survives ±30% perturbation), *over-mechanical agents* (it survives strategic, adaptive agents), and *method self-fulfilment* (a synthetic placebo with known ground truth shows the knockout recovers transmitters and zeroes a planted co-mover). The residual threat we cannot remove is that all causal statements are model-based, in the structural sense, rather than the output of a randomized experiment, which is unavailable for historical crises. The convergence of the four checks and the balance-sheet confirmation is the strongest evidence the setting allows. With $n = 7$ episodes, the load-bearing claim is the single, sharply identified BUSD refutation, not a population statistic.

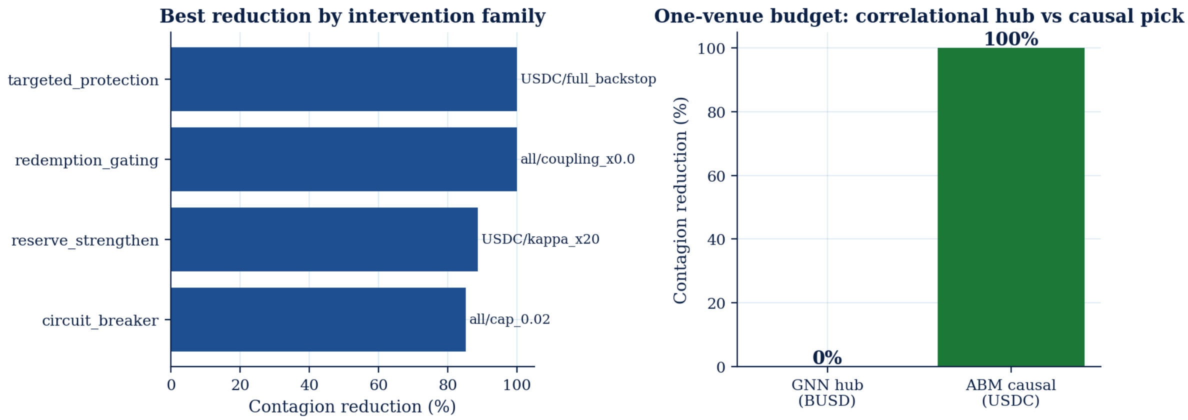

Figure 5: The left panel shows the best contagion reduction by intervention family. The right panel shows a one-venue budget spent on the correlational hub (BUSD) yielding 0%**, and on the causal pick (USDC),** 100%**.**

Three threats to *external* validity deserve a clear statement. First, we hold the transmission network fixed within an episode. In a crisis that ran long enough for venues to actively rewire their reserves, the causal ranking would itself become time-varying, and a single knockout would understate the danger of a node that becomes central only mid-cascade. Second, our calibration targets are second moments (volatility, cross-venue correlation, peg-recovery half-life) rather than the full joint distribution of returns, so a mechanism that leaves those moments unchanged but alters tail behaviour would be invisible to our fit. Relatedly, with five parameters fitted to four moments the calibration is a consistency check rather than an over-identifying test, and the split between panic co-movement (π) and network transmission (c) is identified only through those moments; the BUSD verdict does not rest on this split, since it is analytic (§5). A state-dependent noise extension ($\sigma_{\text{crisis}} = \rho\, \sigma_{\text{calm}}$ for stressed nodes) offers a lever toward tail calibration: the circuit-breaker nonlinearity already yields excess kurtosis $\approx$75, and $\rho \in \{2, 3\}$ tunes it to 14–41 while preserving all four calibrated moments; full kurtosis matching is future work. The near-critical regime we identify is the most this level of calibration can pin down. Third, the balance-sheet network that makes our causal object observable relies on reserve disclosures that are, today, neither universal nor uniformly audited. The very GENIUS Act and MiCA regimes we invoke are what would make this input dependable at scale. None of these overturns the BUSD result, which is identified by the documented reserve structure of the specific episode. They instead mark the boundary of where the method can be trusted without new data.

10 Conclusion

Contagion hub rankings are correlational, and in the marquee SVB crisis the top-ranked hub causes *no* contagion, because, as the balance sheet confirms, no stablecoin's peg depended on it. A calibrated, intervenable model supplies the causal test that centrality cannot, and the disagreement changes the policy outcome. Acting on the correlational hub wastes the entire budget that the causal target would have spent to contain the cascade. As supervisors operationalise the GENIUS Act and MiCA, the practical recommendation is conservative. When the stakes are an intervention, validate a contagion ranking against a causal counterfactual, and where the two disagree, prefer the documented reserve-exposure network: it is transparent, auditable, and computable before any crisis without market data. Contagion analytics intended to guide intervention should report *validated causation*, not centrality alone.

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